Homework #2

Due: 11:59pm, February 2

Reading Assignment:

• Please read BBC (Bresler–Basu–Couvreur) Chapters 4.1, 4.2, 4.4.1, 4.6, and 4.8. (The lecture notes are available in the *Reading Material* section on the Carmen course page.) The concepts of vector spaces, linear operators, and adjoints will be required for the first mini-project. These topics will also be covered in class.

1. **[Fourier transform exercise]**
   The normalized sinc function, the rectangular function, and the triangular function are defined respectively as follows:

$$\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \quad \mathrm{rect}(t) = \begin{cases} 0, & |t| > \frac{1}{2} \\ \frac{1}{2}, & |t| = \frac{1}{2} \\ 1, & |t| < \frac{1}{2} \end{cases}, \quad \mathrm{tri}(t) = \begin{cases} 1 - |t|, & |t| < 1 \\ 0 & |t| \geq 1. \end{cases}$$

   (a) It is known that $\mathrm{rect}(t) * \mathrm{rect}(t) = \mathrm{tri}(t)$. Show that

$$\mathrm{rect}(bt) * \mathrm{rect}(bt) = \frac{1}{b}\mathrm{tri}(bt)$$

   for any $b > 0$.

   (b) Let $a > 0$ be a fixed constant. It is know that

$$\mathcal{F}_{\mathrm{CTFT}}\{\mathrm{sinc}(at)\} = \frac{1}{a}\,\mathrm{rect}\left(\frac{\Omega}{2\pi a}\right).$$

   Then show that

$$\mathcal{F}_{\mathrm{CTFT}}\{\mathrm{sinc}^2(at)\} = \frac{1}{a}\,\mathrm{tri}\left(\frac{\Omega}{2\pi a}\right).$$

   (c) Express the Fourier transform of $x(t) = \cos(10\pi at)\mathrm{sinc}^2(at)$ by using the triangular function.

   (d) Plot the graph of $X(j\Omega)$.

2. **[Parseval-Plancherel identity]**
   Prove

$$\int_{-\infty}^{\infty} x^*(t)y(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_{\mathrm{c}}^*(j\Omega)Y_{\mathrm{c}}(j\Omega)d\Omega.$$

   Hint: Let $z(t) = x^*(t)y(t)$. Use the conjugation property and multiplication property. Note that the left-hand side is the CTFT of $z(t)$ at $\Omega = 0$.

3. **[Recovering a stream of Diracs from Fourier measurements]**
Consider the 2π-periodic continuous-time signal

$$x(t) = \sum_{l=-\infty}^{\infty} \sum_{m=1}^{3} \gamma_m \delta(t - \tau_m - 2\pi l).$$

(a) **Fourier series representation:** Determine a closed-form Fourier series representation of $x(t)$. Specifically, derive an explicit expression for the coefficients $\{a_k\}$ in

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jkt}.$$

(b) **Fourier series of an annihilator:** Define

$$y(t) = (e^{jt} - e^{j\tau_1})(e^{jt} - e^{j\tau_2})(e^{jt} - e^{j\tau_3}).$$

Determine the Fourier series coefficients $\{b_k\}$ in

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jkt}.$$

As part of your derivation, show that

$$b_3 = 1, \quad b_k = 0 \quad \text{for } k \notin \{0, 1, 2, 3\},$$

and express b_0, b_1, b_2 explicitly in terms of τ_1, τ_2, τ_3.

(c) **Fourier-series annihilation relation:** Show that $x(t)y(t) = 0$. Then, using standard Fourier-series properties, conclude that the Fourier coefficients satisfy

$$\sum_{m=-\infty}^{\infty} a_{k-m} b_m = 0, \quad \forall k \in \mathbb{Z}. \tag{1}$$

Indeed, since $\{b_k\}$ is supported on $\{0, 1, 2, 3\}$ and $b_3 = 1$, (1) is rewritten as

$$\sum_{m=0}^{3} a_{k-m} b_m = \begin{bmatrix} a_k & a_{k-1} & a_{k-2} & a_{k-3} \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ 1 \end{bmatrix} = 0, \quad \forall k \in \mathbb{Z}, \tag{2}$$

which is equivalent to

$$\begin{bmatrix} a_k & a_{k-1} & a_{k-2} \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = -a_{k-3}, \quad \forall k \in \mathbb{Z}, \tag{3}$$

(d) **Recovering the unknown locations:** Collect (3) for $k = 3, 4, 5$ and write them as a linear system that determines the vector $[b_0, b_1, b_2]^\mathsf{T}$ in terms of $a_0, a_1, \ldots, a_5$. Show that the values $e^{j\tau_1}, e^{j\tau_2}, e^{j\tau_3}$ correspond to the roots of the cubic polynomial

$$B(z) = b_0 + b_1 z + b_2 z^2 + b_3 z^3.$$

Conclude that the time-shift parameters τ_1, τ_2, and τ_3 can be recovered from the selected Fourier series coefficients $a_0, a_1, \ldots, a_5$ of $x(t)$.